



BOARD QUESTION PAPER : MARCH 2023

MATHEMATICS AND STATISTICS

Time: 3 Hrs.

Max. Marks: 80

General instructions:The question paper is divided into **FOUR** sections.

- (1) **Section A:** Q. 1 contains **Eight** multiple choice type of questions, each carrying **Two** marks.
Q. 2 contains **Four** very short answer type questions, each carrying **one** mark.
- (2) **Section B:** Q. 3 to Q. 14 contain **Twelve** short answer type questions, each carrying **Two** marks. (Attempt any **Eight**)
- (3) **Section C:** Q. 15 to Q. 26 contain **Twelve** short answer type questions, each carrying **Three** marks. (Attempt any **Eight**)
- (4) **Section D:** Q. 27 to Q. 34 contain **Eight** long answer type questions, each carrying **Four** marks. (Attempt any **Five**)
- (5) Use of log table is allowed. Use of calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) Use of graph paper is not necessary. Only rough sketch of graph is expected.
- (8) For each multiple choice type of question, it is mandatory to write the correct answer along with its alphabet, e.g. (a)..... / (b)..... / (c)..... / (d)....., etc. No marks shall be given, if ONLY the correct answer or the alphabet of correct answer is written. Only the first attempt will be considered for evaluation.
- (9) Start answer to each section on a new page.

SECTION – A**Q.1. Select and write the correct answer for the following multiple choice type of questions: [16]**

- i. If $p \wedge q$ is F, $p \rightarrow q$ is F then the truth values of p and q are _____ respectively.
(a) T, T (b) T, F (c) F, T (d) F, F (2)
- ii. In $\triangle ABC$, if $c^2 + a^2 - b^2 = ac$, then $\angle B =$ _____.
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{6}$ (2)
- iii. The area of the triangle with vertices (1, 2, 0), (1, 0, 2) and (0, 3, 1) in sq. unit is _____.
(a) $\sqrt{5}$ (b) $\sqrt{7}$ (c) $\sqrt{6}$ (d) $\sqrt{3}$ (2)
- iv. If the corner points of the feasible solution are (0, 10), (2, 2) and (4, 0) then the point of minimum $z = 3x + 2y$ is _____.
(a) (2, 2) (b) (0, 10) (c) (4, 0) (d) (3, 4) (2)
- v. If y is a function of x and $\log(x + y) = 2xy$, then the value of $y'(0) =$ _____.
(a) 2 (b) 0 (c) -1 (d) 1 (2)
- vi. $\int \cos^3 x \, dx =$ _____.
(a) $\frac{1}{12} \sin 3x + \frac{3}{4} \sin x + c$ (b) $\frac{1}{12} \sin 3x + \frac{1}{4} \sin x + c$
(c) $\frac{1}{12} \sin 3x - \frac{3}{4} \sin x + c$ (d) $\frac{1}{12} \sin 3x - \frac{1}{4} \sin x + c$ (2)
- vii. The solution of the differential equation $\frac{dx}{dt} = \frac{x \log x}{t}$ is _____.
(a) $x = e^{ct}$ (b) $x + e^{ct} = 0$
(c) $x = e^t + t$ (d) $xe^{ct} = 0$ (2)





- viii. Let the probability mass function (p.m.f.) of a random variable X be $P(X = x) = {}^4C_x \left(\frac{5}{9}\right)^x \times \left(\frac{4}{9}\right)^{4-x}$,
for $x = 0, 1, 2, 3, 4$ then $E(X)$ is equal to _____
- (a) $\frac{20}{9}$ (b) $\frac{9}{20}$ (c) $\frac{12}{9}$ (d) $\frac{9}{25}$ (2)

Q.2. Answer the following questions:

[4]

- i. Write the joint equation of co-ordinate axes. (1)
- ii. Find the values of c which satisfy $|\bar{c}\bar{u}| = 3$ where $\bar{u} = \hat{i} + 2\hat{j} + 3\hat{k}$. (1)
- iii. Write $\int \cot x \, dx$. (1)
- iv. Write the degree of the differential equation $e^{\frac{dy}{dx}} + \frac{dy}{dx} = x$ (1)

SECTION – B

Attempt any EIGHT of the following questions:

[16]

- Q.3.** Write inverse and contrapositive of the following statement:

If $x < y$ then $x^2 < y^2$ (2)

- Q.4.** If $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$ is a non singular matrix, then find A^{-1} by elementary row transformations.

Hence write the inverse of $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (2)

- Q.5.** Find the cartesian co-ordinates of the point whose polar co-ordinates are $\left(\sqrt{2}, \frac{\pi}{4}\right)$. (2)

- Q.6.** If $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines and $h^2 = ab \neq 0$ then find the ratio of their slopes. (2)

- Q.7.** If $\bar{a}, \bar{b}, \bar{c}$ are the position vectors of the points A, B, C respectively and $5\bar{a} + 3\bar{b} - 8\bar{c} = \bar{0}$ then find the ratio in which the point C divides the line segment AB. (2)

- Q.8.** Solve the following inequations graphically and write the corner points of the feasible region:
 $2x + 3y \leq 6, x + y \geq 2, x \geq 0, y \geq 0$ (2)

- Q.9.** Show that the function $f(x) = x^3 + 10x + 7, x \in \mathbb{R}$ is strictly increasing. (2)

- Q.10.** Evaluate: $\int_0^{\frac{\pi}{2}} \sqrt{1 - \cos 4x} \, dx$ (2)

- Q.11.** Find the area of the region bounded by the curve $y^2 = 4x$, the X-axis and the lines $x = 1, x = 4$ for $y \geq 0$. (2)

- Q.12.** Solve the differential equation
 $\cos x \cos y \, dy - \sin x \sin y \, dx = 0$ (2)

- Q.13.** Find the mean of number randomly selected from 1 to 15. (2)

- Q.14.** Find the area of the region bounded by the curve $y = x^2$ and the line $y = 4$. (2)





SECTION – C

Attempt any EIGHT of the following questions:

[24]

Q.15. Find the general solution of $\sin \theta + \sin 3\theta + \sin 5\theta = 0$

(3)

Q.16. If $-1 \leq x \leq 1$, then prove that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

(3)

Q.17. If θ is the acute angle between the lines represented by $ax^2 + 2hxy + by^2 = 0$ then prove that

$$\tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right|$$

(3)

Q.18. Find the direction ratios of a vector perpendicular to the two lines whose direction ratios are $-2, 1, -1$ and $-3, -4, 1$.

(3)

Q.19. Find the shortest distance between lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$.

(3)

Q.20. Lines $\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(2\hat{i} - 2\hat{j} + \hat{k})$ and $\vec{r} = (4\hat{i} - 3\hat{j} + 2\hat{k}) + \mu(\hat{i} - 2\hat{j} + 2\hat{k})$ are coplanar. Find the equation of the plane determined by them.

(3)

Q.21. If $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \dots + \infty}}}$, then

$$\text{show that } \frac{dy}{dx} = \frac{\sec^2 x}{2y-1}.$$

Find $\frac{dy}{dx}$ at $x = 0$.

(3)

Q.22. Find the approximate value of $\sin(30^\circ 30')$.Give that $1^\circ = 0.0175^c$ and $\cos 30^\circ = 0.866$

(3)

Q.23. Evaluate $\int x \tan^{-1} x \, dx$

(3)

Q.24. Find the particular solution of the differential equation $\frac{dy}{dx} = e^{2y} \cos x$, when $x = \frac{\pi}{6}$, $y = 0$

(3)

Q.25. For the following probability density function of a random variable X , find (a) $P(X < 1)$ and (b) $P(|X| < 1)$.

$$f(x) = \frac{x+2}{18} \quad ; \text{ for } -2 < x < 4$$

$$= 0 \quad , \text{ otherwise}$$

(3)

Q.26. A die is thrown 6 times. If 'getting an odd number' is a success, find the probability of at least 5 successes.

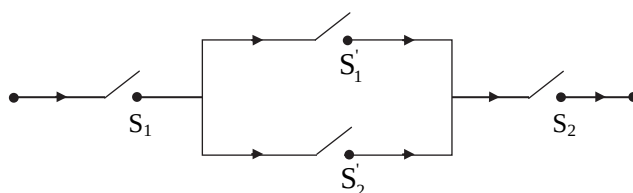
(3)

SECTION – D

Attempt any FIVE of the following questions:

[20]

Q.27. Simplify the given circuit by writing its logical expression. Also write your conclusion.



(4)

Q.28. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ verify that $A(\text{adj}A) = (\text{adj}A)A = |A|I$

(4)





Q.29. Prove that the volume of a tetrahedron with coterminus edges $\vec{a}$, $\vec{b}$ and $\vec{c}$ is $\frac{1}{6}[\vec{a} \vec{b} \vec{c}]$.

Hence, find the volume of tetrahedron whose coterminus edges are $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} + 4\hat{k}$.

(4)

Q.30. Find the length of the perpendicular drawn from the point $P(3, 2, 1)$ to the line $\vec{r} = (7\hat{i} + 7\hat{j} + 6\hat{k}) + \lambda(-2\hat{i} + 2\hat{j} + 3\hat{k})$

(4)

Q.31. If $y = \cos(m \cos^{-1} x)$ then show that $(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + m^2 y = 0$

(4)

Q.32. Verify Lagrange's mean value theorem for the function $f(x) = \sqrt{x+4}$ on the interval $[0, 5]$.

(4)

Q.33. Evaluate: $\int \frac{2x^2 - 3}{(x^2 - 5)(x^2 + 4)} dx$

(4)

Q.34. Prove that: $\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx$

(4)

